

Square $n \times n$ A

$$A \underline{x} = \underbrace{\lambda \underline{x}}_{\lambda I_n \underline{x}}$$

$$\frac{d e^{z x}}{d x} = z e^{z x}$$

$$A \underline{x} - \lambda I_n \underline{x} = \underline{0}$$

$$(A - \lambda I_n) \underline{x} = \underline{0}$$

Determinant of A $\det(A)$ $|A|$

$$\det: \mathbb{R}^{n \times n} \rightarrow \mathbb{R}.$$

$$1. \det \begin{pmatrix} \alpha \underline{x}^T + \beta \underline{y}^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} = \det \begin{pmatrix} -\alpha \underline{x}^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} + \det \begin{pmatrix} -\beta \underline{y}^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix}.$$

$$(\text{Linearity in a row}) = \alpha \det \begin{pmatrix} -\underline{x}^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} + \beta \det \begin{pmatrix} -\underline{y}^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix}$$

$$2. \det \begin{pmatrix} \vdots \\ -\underline{r}_i^T \\ \vdots \\ -\underline{r}_j^T \\ \vdots \end{pmatrix} = - \det \begin{pmatrix} -\underline{r}_i^T \\ \vdots \\ -\underline{r}_j^T \\ \vdots \end{pmatrix} \quad (\text{swapping rows changes sign}).$$

$$\begin{pmatrix} -\underline{r}_1^T \\ -\underline{r}_2^T \\ \vdots \end{pmatrix} \xrightarrow{r_2 \leftarrow r_2 + \alpha r_1} \begin{pmatrix} -\underline{r}_1^T \\ -\underline{r}_2^T + \alpha \underline{r}_1^T \\ \vdots \end{pmatrix}$$

$$\det \begin{pmatrix} -\underline{r}_1^T \\ -\underline{r}_2^T + \alpha \underline{r}_1^T \\ \vdots \end{pmatrix} = \det \begin{pmatrix} -\underline{r}_1^T \\ -\underline{r}_2^T \\ \vdots \end{pmatrix} + \alpha \det \begin{pmatrix} -\underline{r}_1^T \\ \underline{r}_1^T \\ \vdots \end{pmatrix}$$

$$3. \det(I_n) = 1.$$

$$\det \begin{pmatrix} 0 \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} = \det \begin{pmatrix} \underline{r}_2^T - \underline{r}_2^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} = \det \begin{pmatrix} -\underline{r}_2^T \\ -\underline{r}_2^T \\ \vdots \end{pmatrix} - \det \begin{pmatrix} \underline{r}_2^T \\ -\underline{r}_2^T \\ \vdots \end{pmatrix} = 0.$$

Coro Navy:

If A has all pivot cols
If A has a free col

$$\det(A) = \text{prod of pivots.}$$

$$\det(A) = 0.$$

A is invertible iff $\det(A) \neq 0$.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\text{row 1} = a \begin{pmatrix} 1 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \end{pmatrix}$$

$$\text{row 2} = c(1 \ 0) + d(0 \ 1)$$

$$\det(A) = \begin{vmatrix} a(10) + b(01) \\ c & d \end{vmatrix} = a \det \begin{pmatrix} 1 & 0 \\ c & d \end{pmatrix} + b \det \begin{pmatrix} 0 & 1 \\ c & d \end{pmatrix}.$$

$$a \cancel{c} \det \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} + \overset{\text{ad}}{\swarrow} a d \det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \cancel{c} d \det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \cancel{b} d \det \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\frac{ad(+1) + bc(-1)}{= ad - bc.}$$